

# Functional Annotation Report: *fabF* (Q88LL4, PP\_1916) in *Pseudomonas putida* KT2440

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## Gene/Protein Identity Verification

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- **UniProt:** Q88LL4
- **Gene:** *fabF* (ordered locus PP\_1916)
- **Organism:** *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 (PSEPK)
- **Protein:** 3-oxoacyl-[acyl-carrier-protein] synthase 2 ( $\beta$ -ketoacyl-ACP synthase II, "KAS II")
- **EC:** 2.3.1.179
- **Family/domains:** Thiolase-like superfamily;  $\beta$ -ketoacyl synthase N- and C-terminal domains (IPR014030/IPR014031), 3-oxoacyl-ACP\_synth-2 (IPR017568), ketoacyl-synthase active-site signature (IPR018201).

**Verification outcome — CONFIRMED, unambiguous.** The gene symbol *fabF*, the EC number 2.3.1.179, the "3-oxoacyl-ACP synthase 2 / KAS II" description, and the thiolase-fold ketosynthase domain architecture are mutually consistent and all point to the same, well-defined enzyme: the elongating condensing enzyme of bacterial type II fatty acid synthesis (FAS-II). Organism-specific genetic work in *Pseudomonas putida* (F1;

**P** 34181948

directly characterizes the FabF orthologs and confirms the function assigned here. No conflicting gene of the same symbol was encountered.

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## 1. Summary (Answer to the Research Question)

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*fabF* encodes  $\beta$ -ketoacyl-[acyl-carrier-protein] synthase II (KAS II; EC 2.3.1.179), the elongating **condensing enzyme** of the dissociated bacterial type II fatty acid synthesis (FAS-II) pathway. In each elongation cycle it catalyzes a decarboxylative **Claisen condensation** that

joins a growing acyl-ACP thioester to **malonyl-ACP**, extending the fatty-acyl chain by two carbons and producing a  $\beta$ -ketoacyl-ACP,  $\text{CO}_2$  and holo-ACP. It is a soluble **cytoplasmic** enzyme that acts on ACP-tethered substrates. Its physiologically distinctive role is the **elongation of palmitoleate (16:1) to cis-vaccenate (18:1)**, which makes it the key enzyme for **temperature-dependent adjustment of membrane phospholipid acyl-chain composition** (homeoviscous adaptation). In *P. putida*, the housekeeping FabF (FabF1, the PP\_1916 ortholog) is required for cis-vaccenic acid synthesis.

## 2. Primary Function: The Reaction Catalyzed

FabF is the **elongating ketosynthase (KAS II)** of FAS-II. It catalyzes the carbon-carbon bond-forming step that lengthens the acyl chain in every round of elongation:



"The  $\beta$ -ketoacyl-ACP synthase FabF catalyzes the key C-C bond formation step for fatty acid elongation in FAS-II" (Huang et al. 2024, ).

**P** 39175097

Unlike the priming enzyme FabH/KAS III (which uses an acyl-CoA donor to start the pathway), the elongating KAS I/II enzymes use **acyl-ACP as the donor** substrate: "in the elongating KASI and KASII enzymes the donor is an acyl-ACP" (Borgaro et al. 2011, ).

**P** 22017312

The acceptor is always **malonyl-ACP**, whose decarboxylation supplies the driving force for the condensation.

### Substrate specificity

FabF is a broad-range elongation enzyme but with characteristic chain-length preferences. In *E. coli*, "FabF efficiently catalyses the elongation of saturated C14 and unsaturated C16:1 acyl-ACP complexes" (Chen et al. 2022, ).

**P** 36048156

Recent structural/enzymological work shows FabF has a **bimodal catalytic pattern**,

preferentially acting on **C6 and C10 acyl-ACP substrates** by using two hydrophobic cavities and "front", "middle" and "back" door residues to stabilize specific chain lengths (Huang et al. 2024,

P 39175097

The most physiologically important specialized reaction is the elongation of **palmitoleoyl-ACP (16:1)** to **cis-vaccenoyl-ACP (18:1)** (de Mendoza, Ulrich & Cronan 1983,

P 6337151

### 3. Catalytic Mechanism and Active-Site Chemistry

FabF adopts the **thiolase ( $\alpha\beta\alpha\beta$ ) fold** and operates as a homodimer, using a two-step **ping-pong mechanism** with covalent catalysis and a **Cys–His–His catalytic triad**:

1. **Acyl-enzyme formation (transacylation)**: the acyl group of the donor acyl-ACP is transferred to the active-site nucleophilic **cysteine**, forming a covalent acyl-enzyme intermediate. "The nucleophilic cysteine of the active site triad was required for acyl-enzyme formation and the overall condensation activity" (Zhang et al. 2006,

P 16618705

2. **Decarboxylative condensation (Claisen)**: malonyl-ACP binds; one histidine (His337 in *S. pneumoniae* numbering) stabilizes the malonyl thioester oxyanion/transition state, while a second histidine (His303) deprotonates a structured active-site water and promotes decarboxylation of malonate, "releasing bicarbonate"; the resulting carbanion attacks the acyl-enzyme thioester carbonyl to form the new C–C bond, yielding  $\beta$ -ketoacyl-ACP

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Quantum-chemical analysis of FabF assigns the equivalent residues **Lys336 and His304** to "proton transfer during condensation catalysis and C-C bond formation" (Huang et al. 2024,

P 39175097

**Substrate delivery is gated.** Crystal structures of *E. coli* FabF and FabB in complex with substrate-mimetic acyl-ACPs show that "two active site loops undergo large conformational excursions during a dynamic gating mechanism" that "regulates acyl-ACP binding and substrate delivery to the KS active site" (Mindrebo et al. 2020,

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FabF recognizes ACP electrostatically, using positively charged residues on the  $\eta$ 3-helix and loop1 near the catalytic-tunnel entrance to dock the acidic ACP (Huang et al. 2024,

P 39175097

Mechanism-based crosslinking further captured asymmetric, cooperative substrate-binding pockets and a mobile active-site gate residue (Phe) in the related FabB (Chen et al. 2022,

P 36048156

**Pharmacological validation of the mechanism.** The active-site cysteine is the target of the natural product **cerulenin**, which forms a covalent adduct; the crystal structure of *B. subtilis* FabF–cerulenin shows the covalent bond forms at inhibitor carbon C3 with Cys163 (Trajtenberg et al. 2014,

P 24641521

FabF/FabB are also the targets of **platensimycin**, which "works by inhibiting beta-ketoacyl synthases I/II (FabF/B)" (Manallack et al. 2008,

P 18336284

These inhibitors confirm that condensation proceeds through the catalytic cysteine/malonyl-binding pocket.

## 4. Biological Process and Pathway Context

FabF functions within **type II (dissociated) fatty acid synthesis (FAS-II)**, the pathway that supplies the acyl chains for membrane phospholipids and lipid-derived cofactors. Each elongation cycle consists of: condensation (**FabB/FabF**) → reduction (FabG) → dehydration (FabA/FabZ) → enoyl reduction (FabI), all acting on ACP-bound intermediates. FabF performs the **committed elongation/condensation** step of this cycle. The pathway is essential for membrane biogenesis and for producing intermediates used in vitamin/cofactor biosynthesis, and it is a validated antibacterial target (Radka & Rock 2022,

P 35650664

## Specialized role: unsaturated fatty acids and membrane homeostasis

FabF's signature physiological contribution is control of **membrane fluidity via homeoviscous adaptation**. In *E. coli*, "strains (fabF-) lacking beta-ketoacyl-ACP synthase II are deficient in both cis-vaccenic acid synthesis and thermal regulation," and "synthase II, the product of the fabF gene, is the sole enzyme regulating the temperature-dependent composition of the membrane phospholipid acyl chains" (de Mendoza et al. 1983, ).

P 6337151

FabF elongates 16:1 to 18:1 (cis-vaccenate); its activity increases at lower temperature, raising the proportion of the longer, still-unsaturated 18:1 chain to maintain membrane fluidity.

## Pathway role in *Pseudomonas putida* specifically

The soil bacterium *P. putida* has an expanded set of condensing enzymes. In *P. putida* F1, five KAS genes exist — four annotated FabF and one FabB — and genetic dissection shows a clear division of labor: "The *P. putida*  $\Delta$ fabF1 deletion strain was almost entirely defective in synthesis of cis-vaccenic acid," and heterologously expressed *P. putida* fabF1 "restored *E. coli* fabF function" (Dong et al. 2021, ).

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A second paralog, **FabF2, is normally cryptic** (transcriptionally repressed) and only becomes physiologically important when its repressor (Pput\_2425) is inactivated, at which point it can even substitute for FabB in unsaturated fatty acid synthesis ( ).

P 34181948

This establishes **FabF1 — the KT2440 PP\_1916 ortholog — as the primary, housekeeping KAS II elongation enzyme**, while the paralogs provide backup or conditionally specialized activities. This is strong organism-specific support for assigning the classic KAS II function to Q88LL4.

## 5. Subcellular Localization

FabF is a **soluble cytoplasmic enzyme**. In FAS-II, all substrates are tethered as thioesters to the small acidic acyl carrier protein (ACP), and the pathway enzymes — including the condensing enzymes — operate in the **cytosol**, shuttling acyl chains between soluble catalytic domains on

ACP. FabF has no membrane-spanning segment or signal peptide; its "localization" is defined by its function on cytoplasmic ACP-bound acyl intermediates, with the finished acyl chains subsequently transferred to the membrane-associated acyltransferases (PlsB/PlsC/PlsX-PlsY system) for phospholipid assembly. The reliance on protein–protein docking with ACP (via the  $\eta$ 3-helix/loop1 electrostatic interface; Huang et al. 2024,

[P 39175097](#)

is consistent with a cytoplasmic, ACP-dependent enzyme.

## 6. Evidence Summary

| Claim                                                                                     | Evidence type                                | Key reference                                              |
|-------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------|
| FabF catalyzes the C–C bond-forming elongation step of FAS-II                             | Structural + enzymatic (FabF–ACP complexes)  | <a href="#">P 39175097</a>                                 |
| Elongating KAS uses acyl-ACP donor + malonyl-ACP acceptor                                 | Enzymology / substrate mimetics              | <a href="#">P 22017312</a>                                 |
| Ping-pong mechanism, catalytic Cys + two His, decarboxylative condensation                | Mutagenesis / mechanism                      | <a href="#">P 16618705</a>                                 |
| Dynamic gating of acyl-ACP substrate delivery                                             | Crystallography + MD                         | <a href="#">P 32265440</a> ;<br><a href="#">P 36048156</a> |
| FabF elongates 16:1→18:1 and is the sole thermal regulator of membrane acyl chains        | Genetics/physiology ( <i>E. coli</i> )       | <a href="#">P 6337151</a>                                  |
| Housekeeping FabF1 required for cis-vaccenate in <i>P. putida</i> ; multiple KAS paralogs | Gene deletion + heterologous complementation | <a href="#">P 34181948</a>                                 |
| Active-site Cys is covalently modified by cerulenin; FabF is a validated drug target      | Crystallography (FabF–cerulenin)             | <a href="#">P 24641521</a> ;<br><a href="#">P 18336284</a> |

## 7. Supported and Refuted Hypotheses

**Supported** - FabF (Q88LL4/PP\_1916) is a  $\beta$ -ketoacyl-ACP synthase II catalyzing the elongation condensation step of FAS-II (EC 2.3.1.179). - It operates by a ping-pong Claisen mechanism using a Cys–His–His triad and malonyl-ACP decarboxylation. - It preferentially elongates 16:1 to 18:1 (cis-vaccenate) and contributes to temperature-dependent membrane composition. - In *P. putida*, the housekeeping FabF (FabF1 ortholog) is genetically required for cis-vaccenic acid synthesis.

**Refuted / excluded** - FabF is *not* a priming enzyme (that is FabH/KAS III, which uses acyl-CoA and initiates the pathway). - FabF is *not* a membrane-integral protein; it acts on cytoplasmic ACP-bound substrates. - The gene symbol is *not* ambiguous here — all identifiers converge on one enzyme.

## 8. Limitations and Future Directions

- Direct biochemical/structural characterization of the specific KT2440 protein (Q88LL4/PP\_1916) was not found in the literature; functional assignment rests on (i) unambiguous sequence/domain annotation, (ii) closely related *P. putida* F1 genetics ( P 34181948 ), and (iii) extensive mechanistic work on *E. coli*, *B. subtilis* and *S. pneumoniae* orthologs. Given the strong sequence and genetic conservation, this inference is robust.
- The precise in vitro chain-length profile and kinetic parameters of the KT2440 enzyme, and the exact partitioning of elongation among the multiple KAS paralogs under different conditions (temperature, solvent stress), remain to be measured directly.
- Future work: recombinant expression and steady-state kinetics of PP\_1916 FabF with defined acyl-ACP substrates; crystal/cryo-EM structure of the KT2440 enzyme–ACP complex; conditional knockdown to quantify its contribution versus paralogs in KT2440 membrane lipid composition and solvent/stress tolerance.

## References (PMIDs)

6337151, 18336284, 22017312, 24641521, 32265440, 34181948, 35650664, 36048156, 39175097.

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